

SOUTHWEST MICHIGAN REGIONAL, MI, USA

WMO: 726355

Lat: 42.126N

Lon: 86.428W

Elev: 643

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 94871

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.7	(c) 9.6	(d) -2.0	(e) 5.1	(f) 6.3	(g) 1.7	(h) 6.2	(i) 10.4	(j) 27.0	(k) 20.8	(l) 25.0	(m) 22.6	(n) 8.9	(o) 180	(p) 0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.7	(c) 89.6	(d) 74.4	(e) 86.3	(f) 72.6	(g) 83.4	(h) 71.6	(i) 76.8	(j) 85.6	(k) 75.2	(l) 83.0	(m) 73.6	(n) 80.7	(o) 10.6	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.5	127.5	81.4	72.5	123.4	80.2	71.6	119.4	78.9	40.6	85.7	39.0	83.1	37.6	80.8	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.8	(b) 20.1	(c) 18.1	(e) -3.3	(f) 94.0	(g) 7.1	(h) 3.3	(i) -8.4	(j) 96.4	(k) -12.5	(l) 98.3	(m) -16.5	(n) 100.1	(o) -21.6	(p) 102.5		
(4) 23.8	(5) 20.1	(6) 18.1	(7) -3.3	(8) 94.0	(9) 7.1	(10) 3.3	(11) -8.4	(12) 96.4	(13) -12.5	(14) 98.3	(15) -16.5	(16) 100.1	(17) -21.6	(18) 102.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	26.5	28.6	36.4	47.2	57.5	66.8	71.1	69.5	63.4	52.4	41.4	31.8	
	DBStd	17.97	10.50	9.80	10.73	9.46	9.28	7.72	6.43	5.57	7.72	8.74	8.83	9.21	
	HDD50	2902	731	602	445	164	30	1	0	0	2	74	285	568	
	HDD65	6280	1195	1019	887	538	271	70	18	21	122	402	709	1028	
	CDD50	2717	1	2	24	81	263	504	654	603	403	150	27	5	
	CDD65	621	0	0	2	5	39	123	207	159	74	12	0	0	
	CDH74	5334	0	0	8	56	388	1117	1874	1218	584	89	0	0	
	CDH80	1528	0	0	0	8	101	344	597	308	157	13	0	0	
Wind	WSAvg	7.9	10.1	9.4	8.7	8.7	7.3	6.2	5.7	5.3	6.0	7.9	9.2	9.9	
Precipitation	PrecAvg	35.2	2.3	1.8	2.0	3.0	3.4	3.5	3.3	4.0	3.5	3.6	3.1	2.3	
	PrecMax	45.2	4.9	4.3	4.2	5.2	6.7	7.6	7.2	10.9	11.0	8.7	7.2	5.7	
	PrecMin	27.4	0.6	0.5	0.5	0.3	0.7	1.0	0.8	1.5	0.9	0.5	1.1	1.0	
	PrecStd	5.1	1.1	0.9	1.0	1.3	1.7	1.9	1.5	2.2	2.3	2.1	1.5	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	61.5	75.0	80.8	87.9	91.2	93.4	91.0	89.4	81.7	69.6	61.4	
		MCWB	55.2	51.0	62.4	62.9	70.7	74.1	77.0	75.9	73.5	69.4	57.8	57.0	
	2%	DB	50.6	53.9	66.5	74.2	82.5	88.0	89.8	86.8	84.2	75.5	63.4	54.7	
		MCWB	46.7	48.1	57.6	59.9	67.6	72.6	75.8	73.4	71.6	64.1	55.5	50.6	
	5%	DB	44.9	46.2	60.6	69.9	78.6	83.8	86.1	83.2	80.9	71.7	59.6	48.4	
		MCWB	41.1	41.5	53.4	56.7	65.6	71.0	73.6	72.1	69.3	63.2	52.8	44.7	
	10%	DB	39.9	42.2	53.9	64.0	73.7	80.9	82.5	80.7	76.7	66.8	55.0	44.2	
		MCWB	37.3	37.9	47.7	54.1	62.9	69.5	72.4	71.1	67.4	60.1	49.3	40.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.4	56.0	63.4	66.6	73.3	77.2	79.7	78.2	75.6	71.9	60.6	58.4
			MCDB	57.7	59.9	72.8	76.9	85.7	88.1	89.6	87.2	84.4	79.8	65.6	61.2
2%		WB	47.4	48.2	59.1	62.0	69.8	74.9	77.5	75.9	73.3	67.7	57.5	51.6	
		MCDB	49.9	53.3	65.9	70.1	79.3	83.8	86.1	83.6	80.9	73.0	62.5	54.0	
5%		WB	41.1	42.4	53.5	58.6	67.3	72.9	75.7	74.2	71.2	64.2	53.8	45.5	
		MCDB	44.0	46.8	59.0	67.2	75.4	81.1	83.7	80.7	77.7	69.4	58.5	47.5	
10%		WB	37.4	37.8	47.5	54.9	64.5	70.9	74.0	72.6	69.1	60.8	50.1	40.8	
		MCDB	40.0	41.3	53.2	63.8	72.0	78.5	81.1	78.3	75.2	66.7	54.8	44.2	
Mean Daily Temperature Range		5% DB	MDBR	12.1	13.0	16.0	19.6	19.9	19.5	18.7	18.4	19.7	17.5	13.5	11.1
			MCDBR	16.6	19.7	24.3	27.2	24.6	22.2	21.5	21.6	23.0	23.3	19.7	16.6
	MCWBR		14.1	15.9	17.2	16.1	13.3	10.8	10.1	11.0	11.9	14.3	14.3	14.4	
	5% WB	MCDDBR	15.7	18.7	22.2	24.0	21.5	19.1	18.8	17.4	18.5	19.5	17.4	14.9	
		MCWBR	14.0	16.2	16.6	15.9	13.0	11.0	10.1	10.2	11.0	13.9	14.1	14.2	
Clear-Sky Solar Irradiance	taub		0.303	0.326	0.345	0.390	0.419	0.430	0.437	0.434	0.407	0.357	0.321	0.305	
	taud		2.427	2.351	2.370	2.278	2.258	2.276	2.258	2.267	2.307	2.441	2.489	2.468	
	Ebn at Noon		267	277	285	279	272	269	265	262	261	263	259	257	
	Edh at Noon		24	31	34	41	43	42	42	41	36	28	23	22	
All-Sky Solar Radiation	RadAvg		417	671	1125	1503	1727	1967	1997	1715	1383	879	524	351	
	RadStd		43	91	92	128	155	126	92	98	116	98	80	44	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page